Kunal Agrawal

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Professional Summary

AI/Software Engineer building production-grade systems across the stack — from LLM-powered agentic pipelines (RAG, LangGraph, MCP) to full-stack apps with Spring Boot, React, and PostgreSQL. Skilled in hybrid retrieval systems, multi-agent workflows, RESTful APIs, and containerized deployments. SIH 2025 National Finalist.

Education

IMS Engineering College Ghaziabad, India
B.Tech - Computer Science and Engineering — GPA: 7.81 October 2023 – June 2027
Courses: Data Structures and Algorithms, Artificial Intelligence, OOPs, Computer Networks, DBMS, Operating Systems

Projects

Incident Intelligence Agent — AI-Powered Root Cause Analysis **July 2026**
LangGraph, FastMCP, PGvector, LiteLLM, Langfuse, Guardrails AI GitHub

- Architected a production-grade Agentic RAG system to autonomously investigate production incidents, targeting root cause analysis time reduction from **˜45 minutes to under 30 seconds**.
- Engineered a hybrid retrieval pipeline using **pgvector HNSW, PostgreSQL Full-Text Search, Reciprocal Rank Fusion (RRF),** and recency-aware ranking; validated on **150 incidents across 15 microservices**, achieving **0.62–0.78** top-1 cosine similarity.
- Developed a self-correcting multi-agent workflow with **3 FastMCP tools**, citation verification, Guardrails AI, LiteLLM routing, and Langfuse/RAGAS integration for reliable, observable, evidence-backed reports.
- Diagnosed and resolved **6+ production-grade infrastructure issues** through systematic root-cause analysis and containerized the complete **5-service** architecture with Docker Compose for reproducible single-command deployment.

DocMind — Offline Multimodal RAG System **June 2026**
Python, FastAPI, Node.js, Qdrant, Whisper, CLIP, Ollama GitHub

- Built a hybrid retrieval engine fusing Qdrant dense search with BM25 sparse search via Reciprocal Rank Fusion (RRF) — combining 2 embedding models and 2 retrieval strategies for higher recall over single-modal baselines.
- Ingested 3 modalities (PDF, images, audio) in a unified pipeline — Tesseract OCR for image-to-text, Whisper for timestamped audio chunking — enabling cross-modal semantic search across all document types.
- Served offline LLM inference via Ollama (Qwen2.5 / Phi-3) with 0 cloud API calls, containerized via Docker Compose (Qdrant + Ollama) for single-command reproducible deployment on air-gapped systems.
- Selected at SIH 2025 Grand Finale (top ˜300 teams from 50,000+) for solving a real government use case: offline document intelligence for restricted-network environments.

ParkEase — Full-Stack Parking Management Platform **November 2025**
Spring Boot, React.js, PostgreSQL, Render, Vercel GitHub — Live Demo

- Developed 12+ RESTful APIs with normalized schemas, improving query efficiency by ˜30% and achieving ¡250ms response time for location-based search.
- Built the full frontend in React.js and backend in Spring Boot, deploying independently on Vercel and Render with optimized read-heavy query performance for concurrent users.

Technical Skills

- **Programming Languages**: Python, JavaScript, Java
- **LLM & GenAI**: LangChain, LangGraph, RAG Pipelines, RAGAS, HuggingFace, Groq API, OpenAI API, Ollama
- **Retrieval & Vector**: FAISS, Qdrant, ChromaDB, RRF hybrid search, Embedding Models, Semantic Search
- **Agentic AI**: MCP (Model Context Protocol), Tool Calling, ReAct Agents
- **Backend & Databases**: FastAPI, Spring Boot, PostgreSQL, MySQL, MongoDB
- **Tools & Deployments**: Git, GitHub, Render, Vercel, Docker, Postman, VS Code, IntelliJ IDEA

Coding Profiles

- **LeetCode** – 250+ Problems — max: 1523
- **CodeChef** – 200+ Problems — max: 1346 — 2 Star
- **Codeforces** – 120+ Problems — max: 817

Achievements

- Smart India Hackathon 2025 – **National Finalist** (Top 300 out of 50,000+ participating teams).
- Solved **400+ Data Structures and Algorithms problems** covering **Arrays, Stacks, Linked Lists, Queues, Trees, and Recursion** across competitive programming platforms.